LED Current Sink Driver IC

AQ06

6-Channel

General Description

The AQ06 is a compact and white package IC featuring 6-channel input voltage-based low side current sink control driver optimized for LED lighting solution.
The self-protection features include thermal regulation, temperature protection and LED short monitoring output.

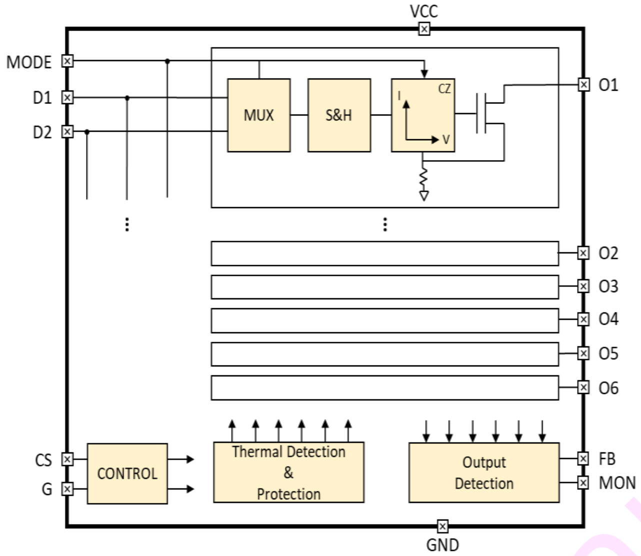
Feature

- All-in-1 package (Dimming control, LED current sink FET, Headroom voltage low detection, LED open & short and thermal-protection mechanism)
- Independent dimming control for each channel in a package
- Maximum 80V headroom voltage for serial connection of multiple LEDs
- Maximum 40 mA sink current per channel for linearity
- 4.5V ~ 5.5V IC operating voltage range
- Multiplexed analog voltage input with gate sampling
- Headroom voltage control signal FB for power supply voltage control
- Fail-safe FB output when LED open detected
- Thermal regulation & over temperature protection
- LED short sensing monitor output
- Low power consumption
- Latch-up Immunity I_{LATCH}: JEDEC JESD78D Nov 2011: ±100mA
- ESD_{HBM} protection (Norm: MIL 883 E Method 3015 Human Body Model): ±2kV
- The optimized pin-out of the AQ06 allows for a dense layout and a compact total solution size
- Small and white color package (QFN2020-16H) for lighting solution

Application

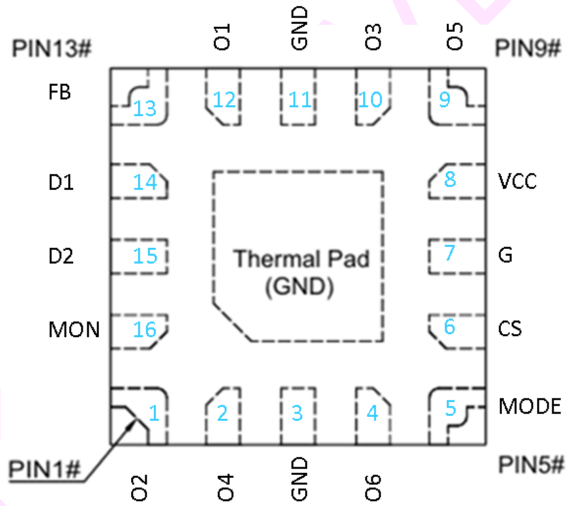
- LED lighting with dimming control

Block Diagram



Pin Diagram

QFN2020-16H
Top View



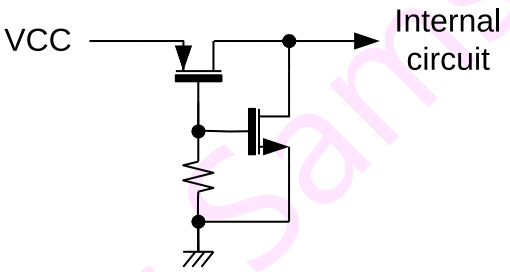
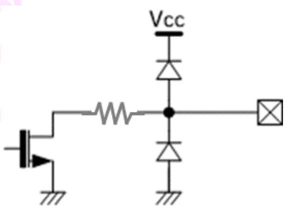
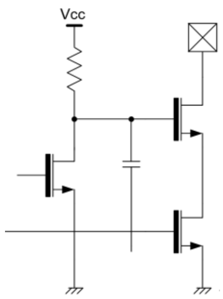
Pin description

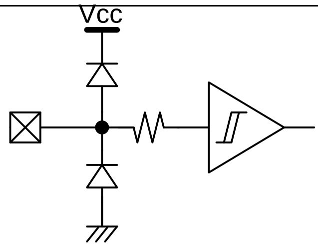
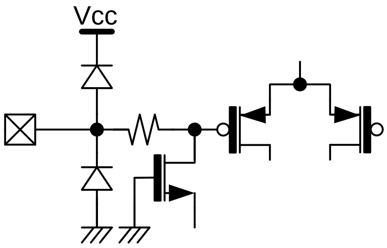
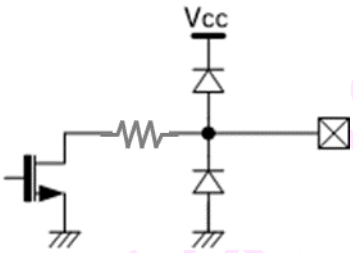
Pin Name	Pin No.	I/O	Type	Description	If pins unused, the levels to be
O2	1	Out	Current load	Current Sink 2	GND 1)
O4	2	Out	Current load	Current Sink 4	GND 1)
GND	3	–	GND	Ground	–
O6	4	Out	Current load	Current Sink 6	GND 1)
MODE	5	In	Digital input	Mode Control	–
CS	6	In	Digital input	Chip Select	–
G	7	In	Digital input	Gate enable	GND
VCC	8	–	Supply voltage	IC operation	+5V (Vcc)
O5	9	Out	Current load	Current Sink 5	GND 1)
O3	10	Out	Current load	Current Sink 3	GND 1)

GND	11	–	GND	Ground	GND
O1	12	Out	Current load	Current Sink 1	GND [1)]
FB	13	Out	Open drain	Headroom voltage low detection	GND
D1	14	In	Analog input	Analog Dimming input 1	GND
D2	15	In	Analog input	Analog Dimming input 2	GND
MON	16	Out	Open drain	Headroom voltage high detection	GND
TPAD	Bottom	–	Thermal Pad	Internally connected to GND	GND

Note: 1) Unused "Ox" (x=1,2,3,4,5,6) output pin connected to GND for reliable operation

Equivalent Circuits of Individual Pins

PAD Name	I/O	Internal Circuit	Description
VCC	–		Supply Voltage input
FB, MON	O		Headroom voltage low detection
Ox	O		Current sink drive output
CS,	I		Gate enable

G			
D, MODE	I		Analog Dimming input
MON	O		Headroom voltage high detection

Maximum Absolute ratings (Ta=25°C unless otherwise specified)

Parameter	Symbol	Value	Unit
Supply voltage	V_{CC}	-0.3 ~ 7.0	V
Maximum output pin voltage ; $Io_{(BR)}$ 1mA	V_O	100	V
MODE, D to GND voltage	V_{MODE}, V_D	-0.3 ~ Vcc+0.3	V
CS, G to GND voltage	V_{CS}, V_G	-0.3 ~ Vcc+0.3	V
FB to GND voltage	V_{FB}	-0.3 ~ Vcc+0.3	V
MON to GND voltage	V_{MON}	-0.3 ~ Vcc+0.3	V
FB to GND current	I_{FB}	5	mA
MON to GND current	I_{MON}	5	mA
Thermal Resistance (Junct. to Amb.)	$R_{\theta JA}$	300 2)	℃/W
Thermal Resistance (Junct. to Case)	$R_{\theta JC}$	3 2)	℃/W
Power Dissipation on package 2)	P_{TOT}	1	W
Junction temperature	T_J	-20 ~ 150	℃
Storage temperature	T_{STG}	-45 ~ 150	℃
Relative Humidity(non-condensing)	RH_{NC}	5 ~ 85	%
Moisture Sensitivity	MSL	Maximum floor life time of 168h	3

Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device

Note: 2) Surface mounted on FR-4 substrate PC Board 1” x 1” inch 2-oz copper plate

Electrostatic Discharge

Parameter	Symbol	Value	Unit	Remark
Electrostatic Discharge HBM ; Contact	ESD_{HBM}	±2k	V	C=100 pF, R=1.5 kΩ
Electrostatic Discharge CDM ; Contact	ESD_{CDM}	±500	V	R=1 Ω

Electrical Characteristics

VCC = 5.0V, Ta=25.0℃ Unless otherwise noted, x=1,2,3,4

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Supply						
Operating voltage	V_{CC} 3)		4.5	–	5.5	V
Quiescent current	I_{Q_5V}	Dx=0V, Gx=0V Vhd=0.7V, Vcc=5V	–	–	300	uA
Operating Current at MODE 4	I_{CC4}	O1=O2=O3=O4=O5=O6=32mA	–	–	7	mA
Operating Current at MODE 3	I_{CC3}	O1=O2=O3=O4=O5=O6=24mA	–	–	5.3	mA
Operating Current at MODE 2	I_{CC2}	O1=O2=O3=O4=O5=O6=16mA	–	–	3.7	mA
Operating Current at MODE 1	I_{CC1}	O1=O2=O3=O4=O5=O6=8mA	–	–	2	mA
Input						
Data Input voltage	V_D 4)	Dx: Analog input for dimming control	0.23	–	4.5	V
Data hold time	t_{DH}	25°C; Io=20mA → Io=19.5mA	1	–	–	Sec
Input current	I_D, I_{MODE} I_{CS}, I_G	MODE, D=3.5V CS, G=5V	0	–	1	uA
Cs, G	V_{IL}	Logic low level	–	–	1.0	V
	V_{IH}	Logic high level	1.7	–	–	V
	V_{IHYS}	Vcc=5.0V	20m	0.6	–	V
Gate enable Time	t_G 5)	25°C; D:(30mA)	3	–	–	uS

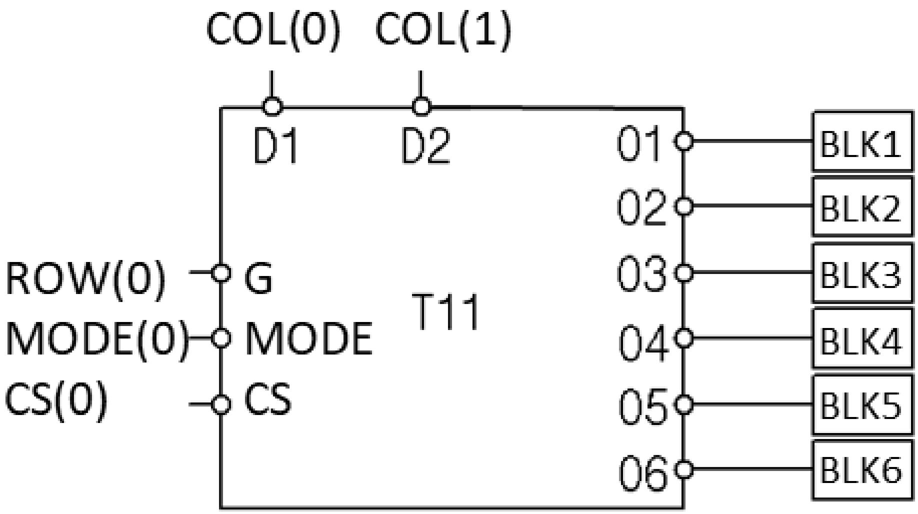
Output						
Current load voltage	V_O	Headroom voltage for normal operation	0.23	–	80	V
Headroom voltage	V_{HD30}	I_{OX}=32mA	450	–	–	mV
Current load off current	I_{OFF1}	D=0V, Gx=0V, Vox=80V			200	nA
Output Current max.	$I_{MODE4\sim1}$	40 mA, 32 mA, 24 mA, 16 mA	16		40	mA
Output Current Accuracy	$Acc_{IO\%}$	400uA @ MODE 4 100uA @ MODE 1	3%	–	+3%	
Control						
"FB active" threshold voltage	V_{FBA}	Logically ORed for 6-channels	630	700	770	mV
"FB release" threshold voltage	V_{FBR}	Logically ORed for 6-channels	670	760	880	mV
"FB" sensing hysteresis voltage	V_{FBH}	Logically ORed for 6-channels	40	60	110	mV
"FB release" by LOE	V_{LOE}	By LED Open Error	140	200	250	mV
FB sink voltage	V_{FBL}	I_{FB}= 2mA (Open Drain)	–	–	0.2	V
MON sink voltage	V_{MON}	I_{MON}=2mA (Open Drain)	–	–	0.2	V
FB detect delay	T_{FBD}	High – > Low	–	–	1	uS
FB low duration	T_{FBHLD}	Minimum low duration	12	–	–	uS
MON low output	T_{MONL}	Synchronized to Gate input high pulse width	–	–	–	uS
Protection and Detection						
Automatic Thermal Regulation Range @T_j	T_{ATR}	Io@125℃ ~ Io/3@150℃	125	–	≈150	℃
Thermal shut down temperature @T_j	T_{OTP}	I_O off	–	150	–	℃
Thermal shut down hysteresis @Tj	T_{HYS}	–	24	30	36	℃
LED short detect voltage threshold	V_{LSD} 6)		26	32	38	V
LED short release voltage threshold	V_{LSR}		23	28	34	V

Note:
3) The operation at Vcc below 5V will limit a maximum LED current sinking.
4) The D input voltage above 3.5V may not guarantee a good linearity in current sinking.
5) Based on internal circuit simulation result.
6) LED short detection logic circuit will not cut off LED current sinking when it is detected.

ATR (Automatic Thermal Regulation) and OTP (Over Temperature Protection)

The junction temperature of the IC must not exceed the maximum limit. Overtemperature protection is activated at 150°C, causing the IC to stop regulating immediately. Overheat protection helps prevent catastrophic failure due to accidental device overheating. If the junction temperature drops below 150°C, the channel will still stop regulating the LED current. When the junction temperature drops below 125°C (typical hysteresis = 30°C), the IC resumes current driving.

Connection Diagram (Single LED block and pin recommendation for display matrix)



Output Current Control by “MODE” Signal Level

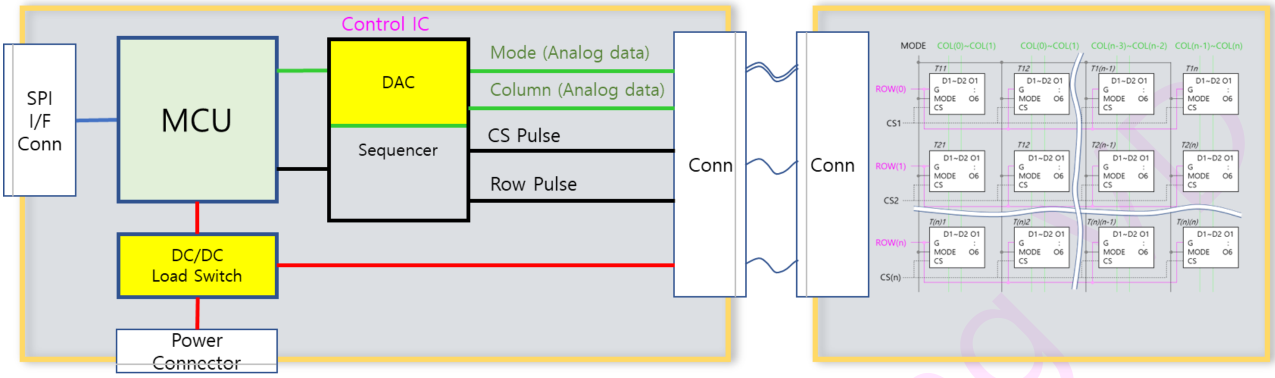
For each output port of the AQ06 IC, the gain of converting the D input voltage to the output current can vary depending on the voltage input level of the MODE pin.

Max. Drive Current Control Table

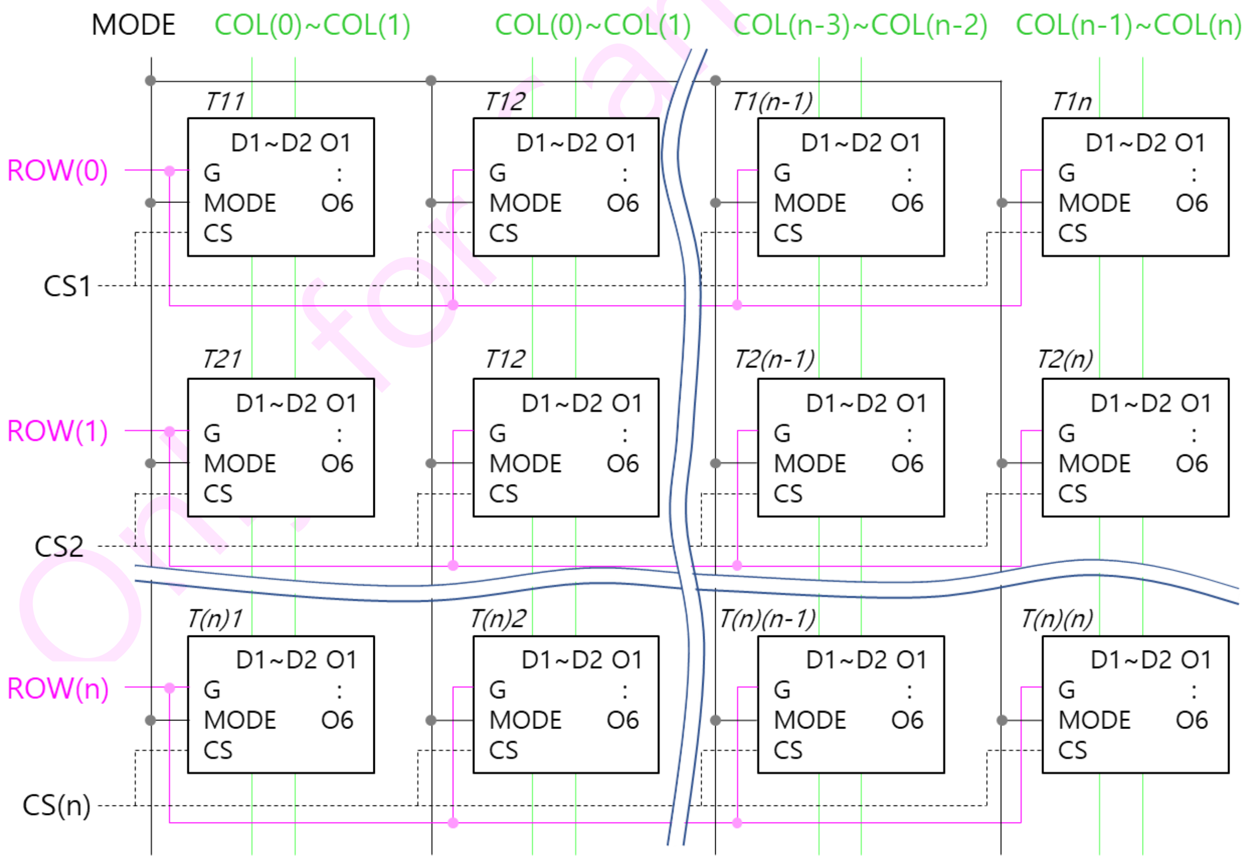
Max Current[1)]	MODE1	MODE2	MODE3	MODE4
16 mA	4 mA	8 mA	12 mA	16 mA
24 mA	6 mA	12 mA	18 mA	24 mA
32 mA	8 mA	16 mA	24 mA	32 mA
40 mA	10 mA	20 mA	30 mA	40 mA

NOTE: 1) The maximum current is determined during the manufacturing process of the IC.

BLU System Block Diagram

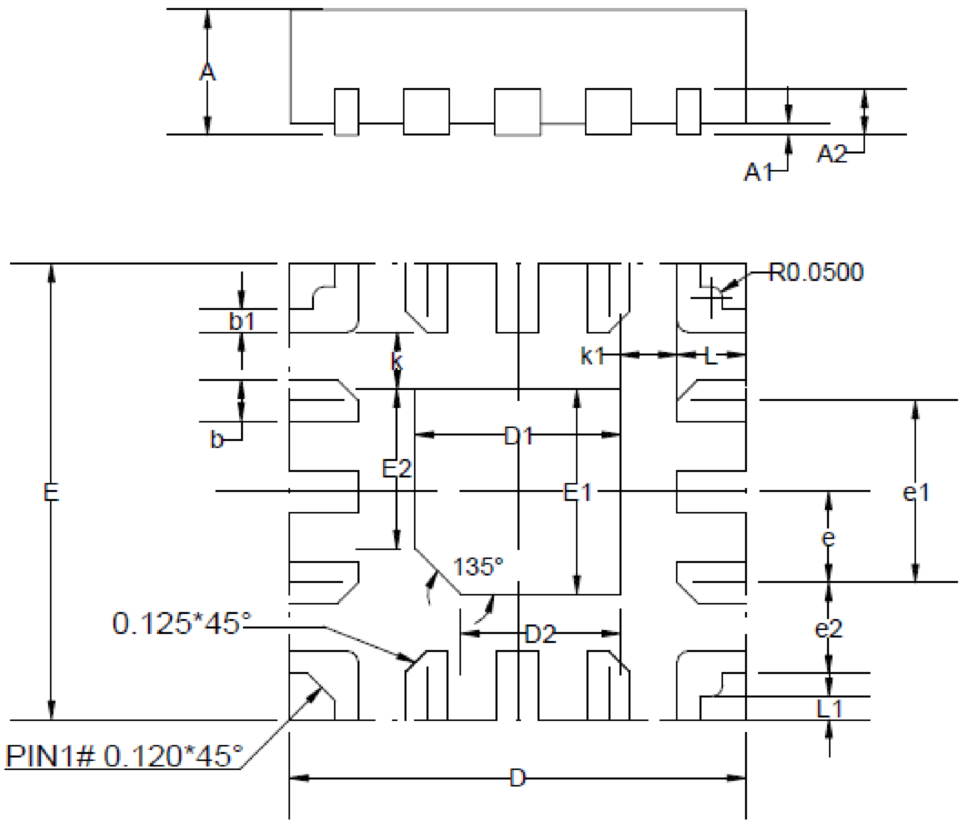


BLU Block Circuit Diagram Around AQ06



Physical Dimensions

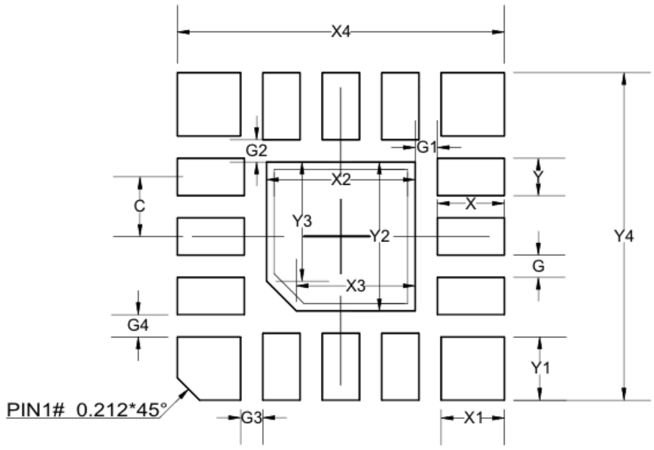
QFN2020-16H Package Outline



UNIT	A	A1	A2	D	D1	D2	E	E1	E2	e	e1	e2
mm	0.45 0.60	0.00 0.05	0.15 Typ.	1.95 2.05	0.85 0.95	0.70 Typ.	1.95 2.05	0.85 0.95	0.70 Typ.	0.40 BSC	0.80 BSC	0.40 BSC

UNIT	L	L1	b	b1	k	k1						
mm	0.25 0.35	0.10 Typ.	0.13 0.23	0.10 Typ.	0.25 BSC	0.25 BSC						

Suggested Footprint – QFN2020-16H



Dimensions	Value(mm)
X	0.45
Y	0.25
C	0.40
X1	0.425
Y1	0.425
X2	1.00
Y2	1.00
X3	0.80
Y3	0.80
X4	2.20
Y4	2.20
G	0.15
G1	0.15
G2	0.15
G3	0.15
G4	0.15

Note: The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD-020 "Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices"